ROLL NO : 241801038

EX: 7 – Sliding Window Protocol

AIM:

To write a Python program to implement flow control at the Data Link Layer using the Sliding Window Protocol and simulate the transmission of frames between two nodes.

ALGORITHM:

1. Start the program.
2. Read the total number of frames (N), window size (W), and the frame number to be simulated as lost.
3. Initialize base = 0 and next_seq = 0.
4. Display the current sender window.
5. Send frames from Node A to Node B within the window size.
6. If a frame is lost, the receiver discards subsequent out-of-order frames.
7. The sender times out and retransmits the lost frame and the remaining frames using the Go-Back-N protocol.
8. If no frame is lost, the receiver accepts the frames and sends acknowledgements (ACKs).
9. Slide the sender window forward after receiving the acknowledgements.
10. Repeat until all frames are successfully transmitted and acknowledged.
11. Display the successful transmission message.
12. Stop the program.

```
def simulate(total_frames, window_size, lost_frame):
    base = 0
    next_seq = 0

    while base < total_frames:
        window_end = min(base + window_size - 1, total_frames - 1)

        print("\n-----------------------------------------")
        print(f"Sender Window: [Frame {base} - Frame {window_end}]")
```

```
        while next_seq < base + window_size and next_seq < total_frames:
            print(f"Node A --> Sending Frame {next_seq}")

            if next_seq == lost_frame:
                print(f"           (Frame {next_seq} LOST in transit!)")

            next_seq += 1

        if lost_frame is not None and base <= lost_frame < next_seq:
            print(f"Node B --> Frame {lost_frame} missing.")
            print("Node B --> Discarding out-of-order frames.")
            print(f"Node A --> Timeout! Resending from Frame {lost_frame}.")

            next_seq = lost_frame
            base = lost_frame
            lost_frame = None
            continue

        for f in range(base, next_seq):
            print(f"Node B --> Received Frame {f}, sending ACK {f}")

        print(f"Node A --> ACK {next_seq - 1} received.")
        print("Node A --> Sliding window forward.")

        base = next_seq

    print("\n-----------------------------------------------")
    print(f"All {total_frames} frames successfully transmitted from Node A to N


def main():
    print("=== SLIDING WINDOW PROTOCOL (GO-BACK-N) ===\n")

    total_frames = int(input("Enter total number of frames: "))
    window_size = int(input("Enter sender's window size: "))
    lost_frame = int(input("Enter lost frame number (-1 for no loss): "))

    if total_frames <= 0 or window_size <= 0:
        print("Please enter positive values.")
        return

    if lost_frame < -1 or lost_frame >= total_frames:
        print("Invalid lost frame number.")
        return

    if lost_frame == -1:
        lost_frame = None

    simulate(total_frames, window_size, lost_frame)
```

```
if __name__ == "__main__":
    main()
```

```
=== SLIDING WINDOW PROTOCOL (GO-BACK-N) ===

Enter total number of frames: 8
Enter sender's window size: 4
Enter lost frame number (-1 for no loss): 5

-----------------------------------------
Sender Window: [Frame 0 - Frame 3]
Node A --> Sending Frame 0
Node A --> Sending Frame 1
Node A --> Sending Frame 2
Node A --> Sending Frame 3
Node B --> Received Frame 0, sending ACK 0
Node B --> Received Frame 1, sending ACK 1
Node B --> Received Frame 2, sending ACK 2
Node B --> Received Frame 3, sending ACK 3
Node A --> ACK 3 received.
Node A --> Sliding window forward.

-----------------------------------------
Sender Window: [Frame 4 - Frame 7]
Node A --> Sending Frame 4
Node A --> Sending Frame 5
           (Frame 5 LOST in transit!)
Node A --> Sending Frame 6
Node A --> Sending Frame 7
Node B --> Frame 5 missing.
Node B --> Discarding out-of-order frames.
Node A --> Timeout! Resending from Frame 5.

-----------------------------------------
Sender Window: [Frame 5 - Frame 7]
Node A --> Sending Frame 5
Node A --> Sending Frame 6
Node A --> Sending Frame 7
Node B --> Received Frame 5, sending ACK 5
Node B --> Received Frame 6, sending ACK 6
Node B --> Received Frame 7, sending ACK 7
Node A --> ACK 7 received.
Node A --> Sliding window forward.

-----------------------------------------
All 8 frames successfully transmitted from Node A to Node B.
```

Result :

The program to implement flow control at the Data Link Layer using the Sliding Window Protocol (Go-Back-N) was successfully written and executed in Python. The

transmission of frames between Node A and Node B, including frame loss detection, retransmission, and acknowledgement, was simulated successfully.